Assignment 2: Simple Search Engine

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1. Methodology

1.1 Overview

The system implements a **naive textual search engine** with three stages: (1) **data preparation** in PySpark, (2) **offline indexing** with Hadoop MapReduce into HDFS and Cassandra, (3) **online retrieval** with PySpark RDDs and **BM25** ranking. Documents are plain UTF-8 text; queries are tokenized consistently with the indexer.

1.2 Data collection and preparation

- **Source:** Wikipedia parquet (id, title, text); categories dropped.
- **Sample:** At least 1,000 articles (configurable in prepare_data.py).
- **HDFS /data:** One file per document named <doc_id>_<doc_title>.txt (spaces in titles become underscores in filenames).
- **HDFS /input/data:** Built with **PySpark** wholeTextFiles("/data/*.txt"), parsing filename for doc_id and title; one line per document:
  doc_id \t doc_title \t doc_text

1.3 Indexing (MapReduce only)

Two **Hadoop Streaming** pipelines (Python mappers/reducers), no Spark in this phase.

Pipeline 1 (mapper1.py / reducer1.py):

- **Input:** /input/data (tab-separated lines).
- **Tokenization:** Lowercase alphanumeric tokens via [a-z0-9]+ on title + body.
- **Map output:**
  - Per document: key DL:<doc_id>, value = document length in tokens.
  - Per (term, doc): key POST:<term>, value = doc_id \t tf (tf from a local count per document).
- **Reduce output (1 reducer):**
  - DOC_LEN \t doc_id \t dl — BM25 document length.
  - VOCAB \t term \t df — document frequency.
  - POST \t term \t doc_id \t tf — inverted index postings.

Pipeline 2 (mapper2.py / reducer2.py):

- **Input:** Output of pipeline 1 (only DOC_LEN lines used).
- **Output:** GLOBAL \t N \t dl_avg \t sum_dl for corpus-level BM25 statistics.

HDFS layout under /indexer:

- pipeline1/, pipeline2/ — full pipeline outputs.
- vocabulary/, index/, stats/doc_lengths/, stats/corpus/ — split views for loading.

Script: create_index.sh [hdfs_input_dir] (default /input/data).

1.4 Storing the index (Cassandra)

Tool: cassandra-driver in store_index.py, invoked by store_index.sh. Data is read from HDFS with hdfs dfs -text on the part files and inserted with batched concurrent writes.

Keyspace: search_index (SimpleStrategy, RF=1).

Table	Primary key	Purpose
vocabulary	term	df for BM25 IDF
postings	(term, doc_id)	tf for each posting
doc_stats	doc_id	dl (length in tokens)
corpus_stats	id (global)	n_docs, dl_avg, sum_dl

Script: index.sh runs create_index.sh then store_index.sh.

1.5 Ranking and retrieval (PySpark)

Application: query.py

- Loads **corpus stats**, **vocabulary** (df for query terms), **postings**, and **doc_stats** from Cassandra (driver).
- Loads **titles** from HDFS /input/data via hdfs dfs -text (driver) to print human-readable titles.
- **BM25** (assignment form, natural log for IDF): for each query term (t), (\text{idf}(t) = \log\frac{N}{\text{df}(t)}),
  term score (\text{idf} \cdot \frac{(k_1+1),\text{tf}}{k_1((1-b)+b,\text{dl}/\text{dl}_{\text{avg}})+\text{tf}}) with (k_1=1), (b=0.75).
  Repeated query terms are weighted with a **Counter** (multiple occurrences add multiple contributions).
- **Spark RDD:** parallelize(postings) → map (per-posting BM25 contribution) → reduceByKey by doc_id → takeOrdered(10) by descending score.

- **Output:** Top 10 lines: doc_id \t title.

Script: search.sh — sets SEARCH_QUERY from **arguments** or **stdin**, runs spark-submit (default --master yarn; local[*] used in docker compose when COMPOSE_SPARK_MASTER is unset for reliability).

1.6 Optional incremental index

add_to_index.sh adds one local <id>_<title>.txt to HDFS /data, rebuilds /input/data with rebuild_input_from_hdfs.py, and runs index.sh (full reindex and Cassandra reload).

2. Demonstration

2.1 How to run the repository

Prerequisites: Docker, Docker Compose; place **app/a.parquet** in the app folder or any other parquet from source but needed to rename it manually in prepare_data.sh.

docker compose up

The master runs app.sh: SSH → Hadoop services → wait for Cassandra → venv + pip install → venv-pack → prepare_data.sh → index.sh → search.sh (default Spark mode local[*] via compose unless COMPOSE_SPARK_MASTER=yarn).

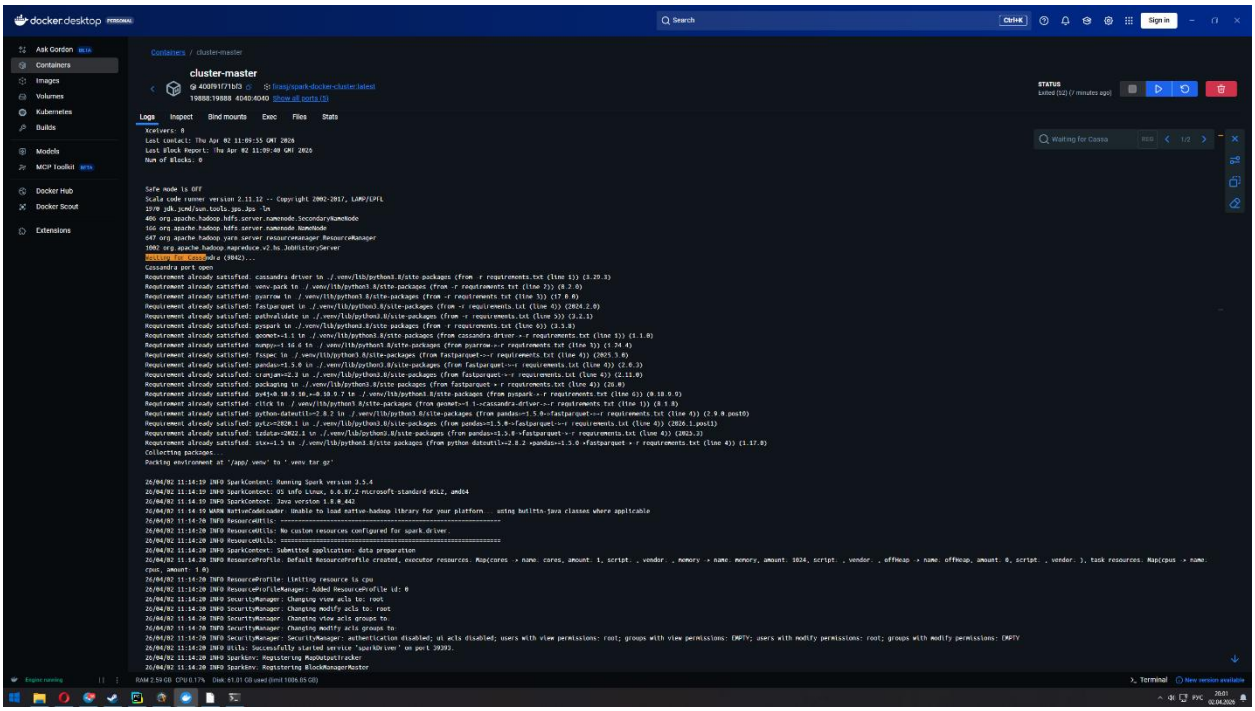
Manual search (after the stack is up):

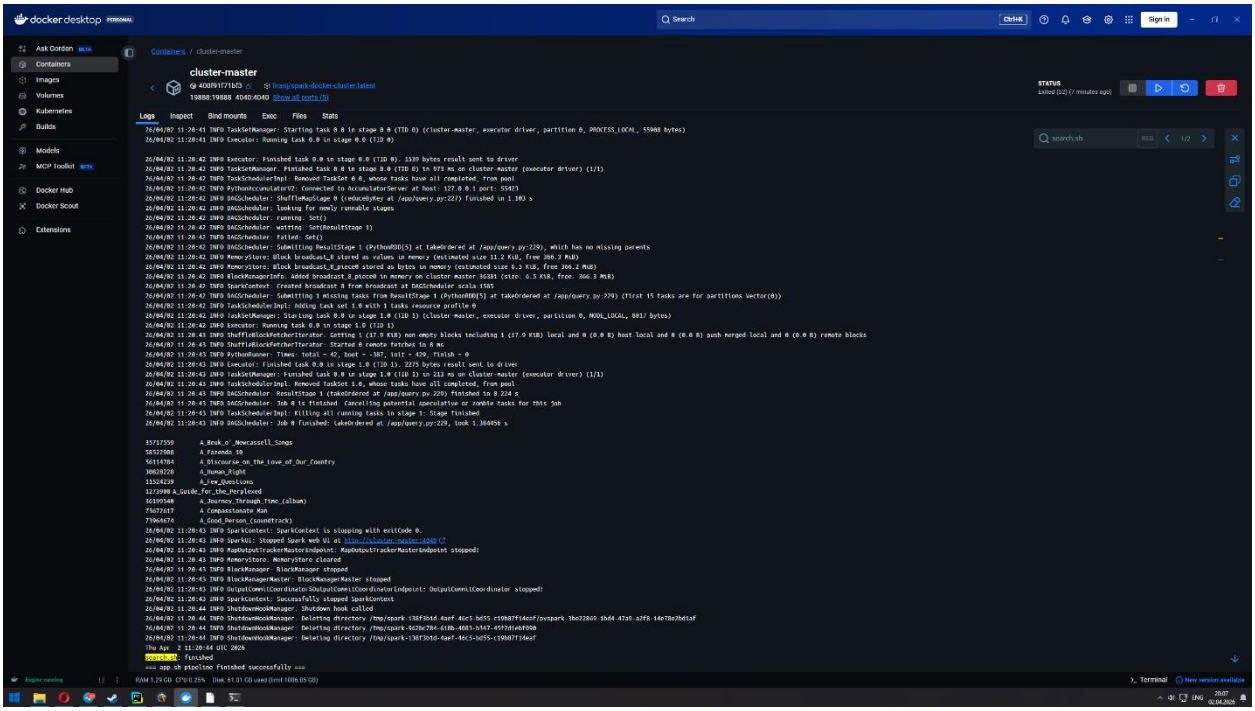
docker exec -it cluster-master bash -lc "cd /app && source .venv/bin/activate && bash search.sh film music wikipedia"

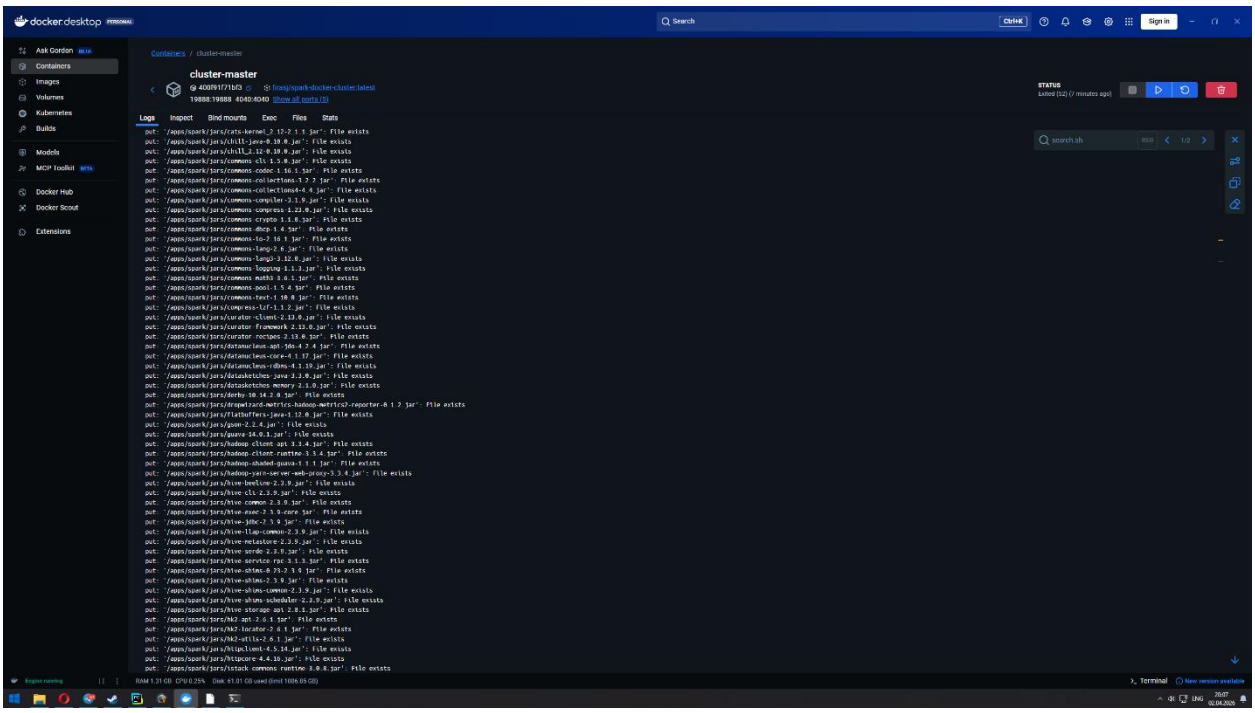
Stdin query:

docker exec -it cluster-master bash -lc "cd /app && source .venv/bin/activate && echo 'history science' | bash search.sh"

2.2 Screenshots



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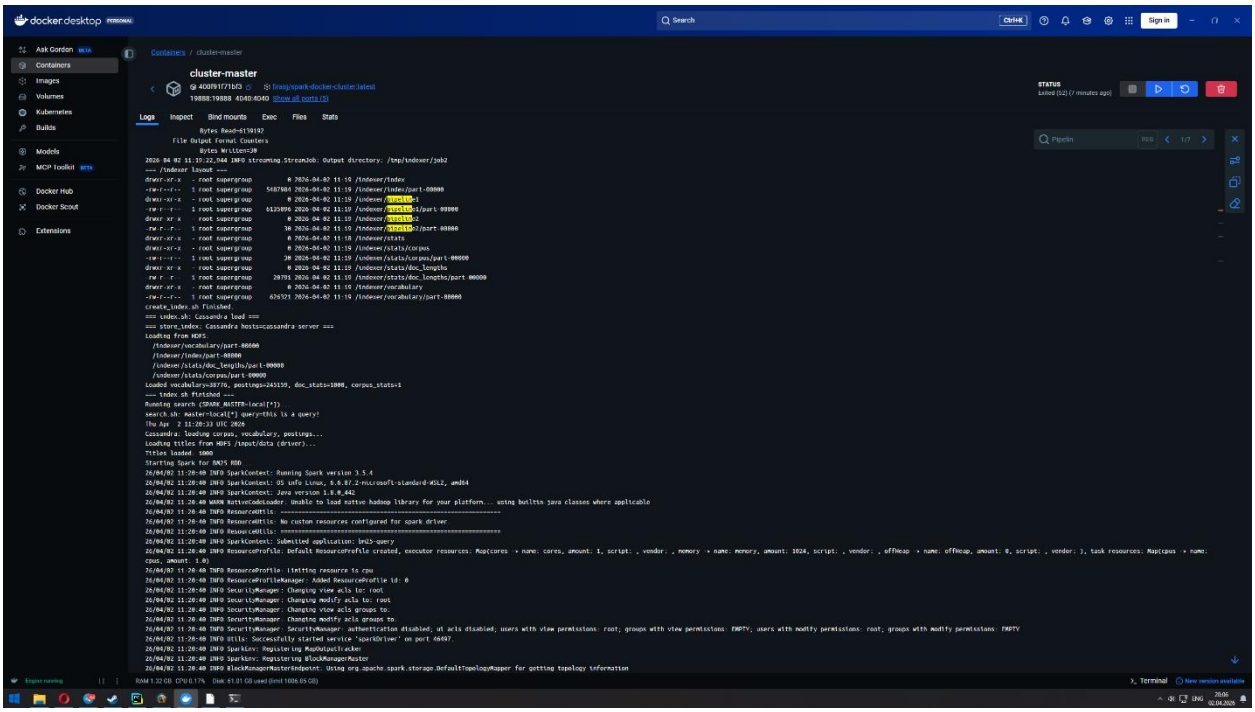
Containers / cluster-master
cluster-master
400f91f71bf3
finaq/spark-docker-cluster:latest
19888:19888 4040:4040
Logs
Inspect
Bind mounts
Exec
Files
Stats
STATUS
Exited (12) (7 minutes ago)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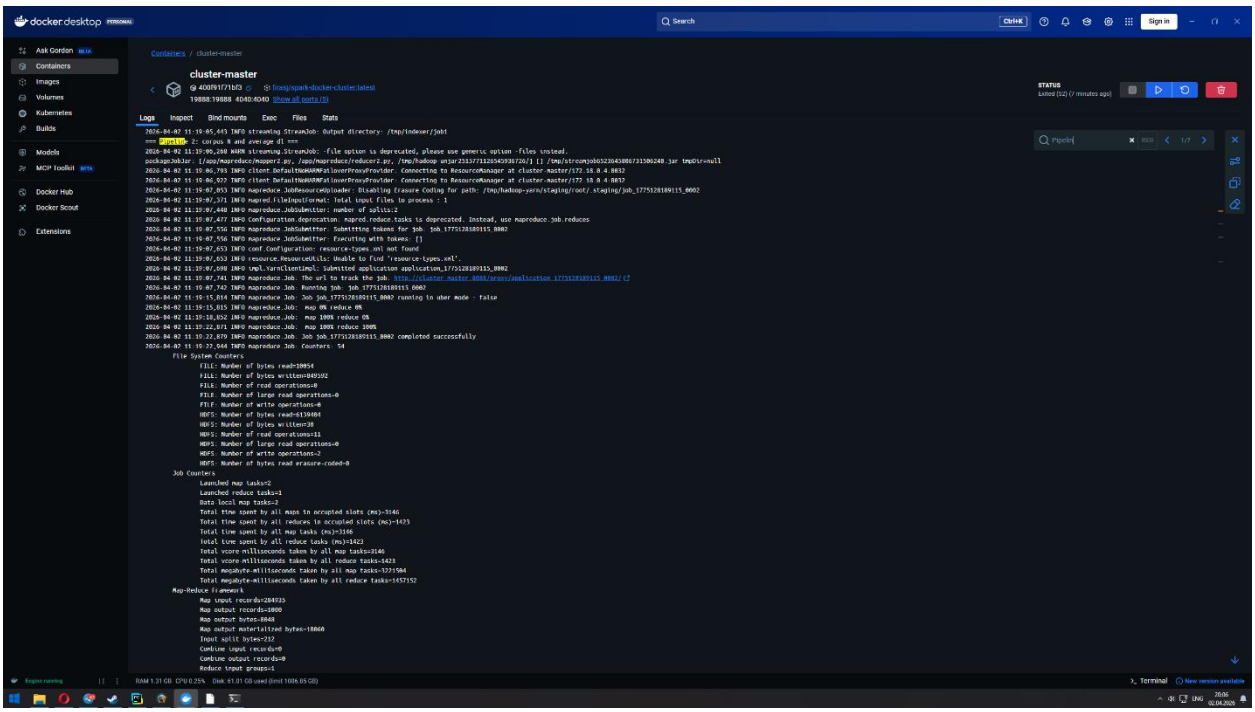
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Ask Gordon
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit
Docker Hub
Docker Scout
Extensions
Containers / cluster-master
cluster-master
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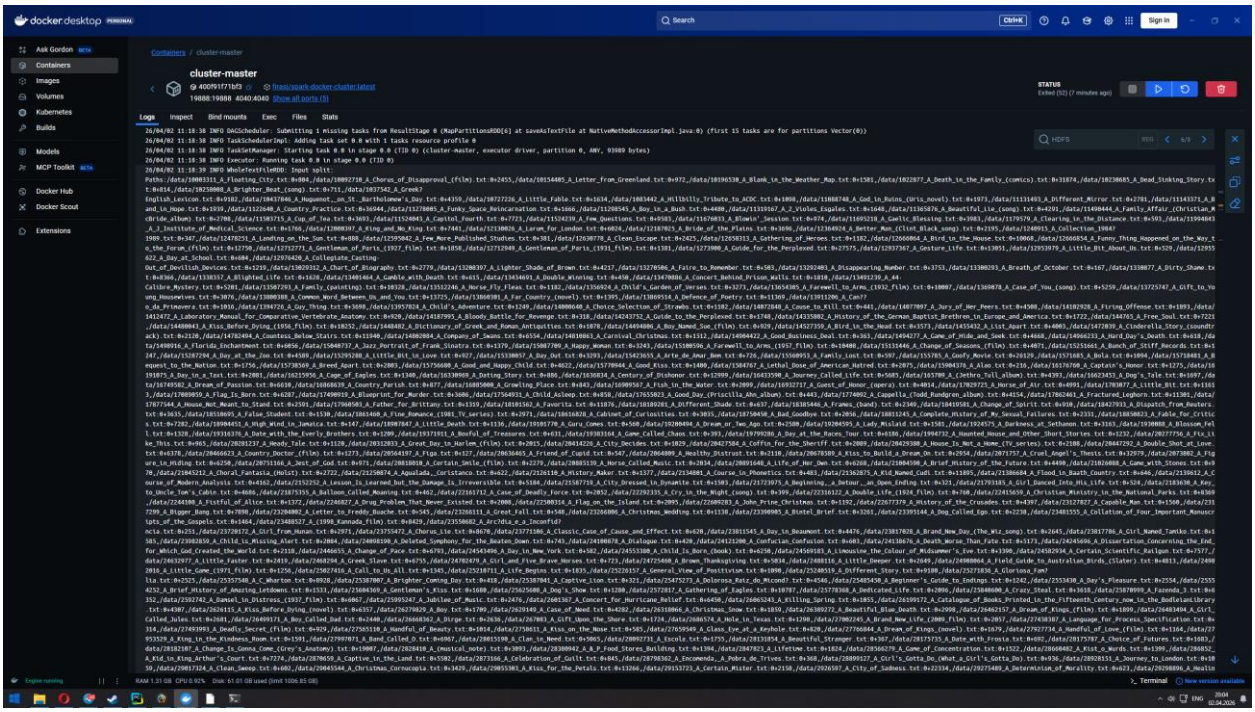
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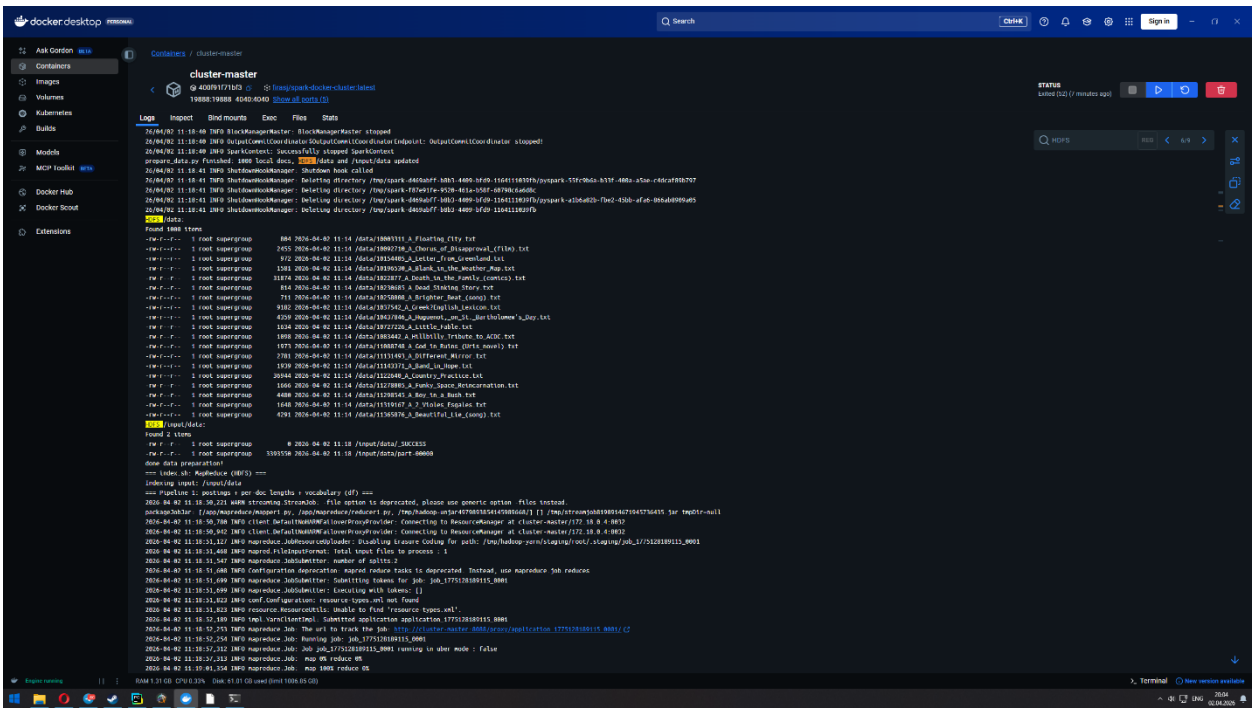
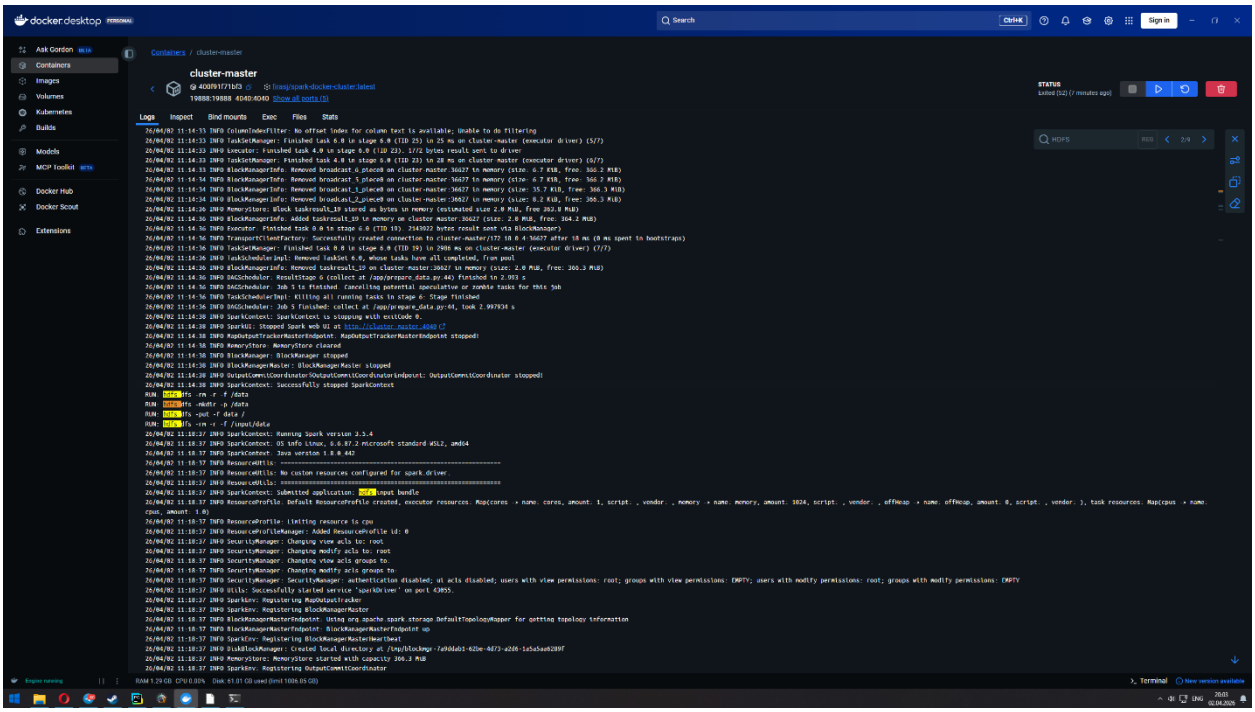

docker desktop
Containers / cluster-master
cluster-master
400f91f71bf3
finaj/spark-docker-cluster:latest
19888:19888 4040:4040
Logs
Inspect
Bind mounts
Exec
Files
Stats
STATUS
Exited (12) (7 minutes ago)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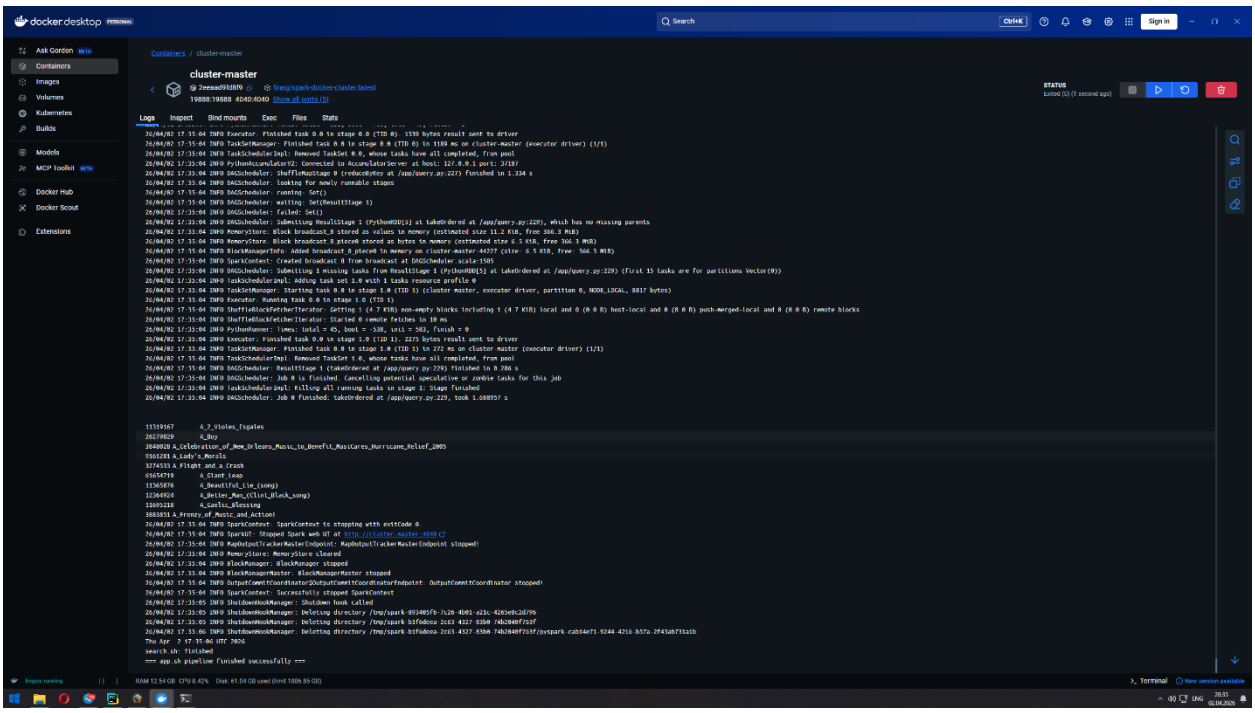
Exited (12) (7 minutes ago)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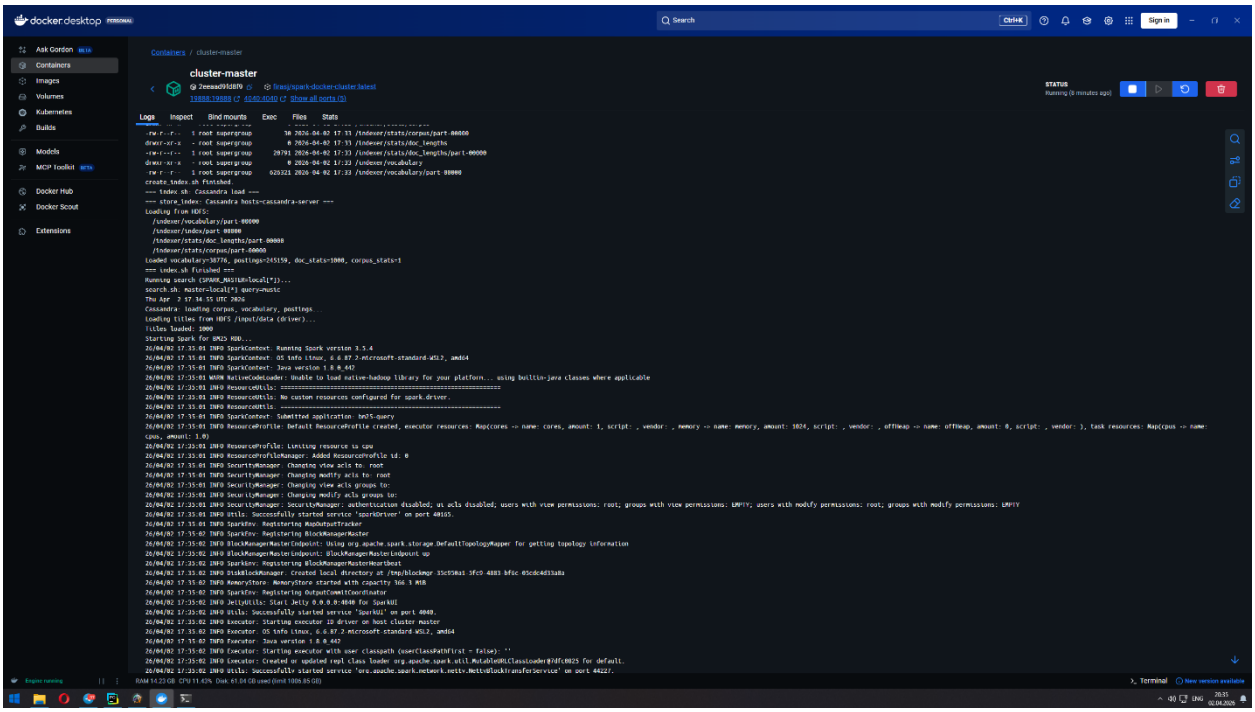
Logs
Inspect
Bind mounts
Exec
Files
Stats
STATUS
Exited (0) (1 second ago)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Containers / cluster-master
cluster-master
2eeaad9fd8f9
firasj/spark-docker-cluster:latest
19888:19888 4040:4040
Logs
Inspect
Bind mounts
Exec
Files
Stats
STATUS
Running (8 minutes ago)